Artun Balta

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Profile

Electrical and electronics engineer working at the intersection of neural signal processing and applied machine learning. Incoming MRes Neurotechnology researcher at Imperial College London, decoding auditory attention from single-neuron recordings, and co-founder of a funded AI voice-agent company with a shipped product and five clinic pilots.

Education

Imperial College London · *MRes Neurotechnology (incoming)* London, UK · 2026 – 2027

- Thesis (supervisor: Dr. Andriy Kozlov): *Auditory Attention Decoding, Adapting Dynamic Spectrotemporal Receptive Fields to Single-Neuron Spike Recordings.* Extends deep-network dSTRF estimation from population-level ECoG to single-unit spike trains, introducing Poisson likelihood formulations and sparsity handling to characterise nonlinear tuning in the auditory pathway.

İhsan Doğramacı Bilkent University · *B.Sc. Electrical and Electronics Engineering* Ankara, Turkey · 2021 – 2026

- **1st Place** (12 teams), Graduation Project Competition · **2nd Place** (45 teams), Bilkentpreneurs Challenge 2026, $3,000 in prize funding · Bilkent Entrepreneurship Award.

Lycée Tevfik Fikret · *High School, Mathematics & Sciences* Ankara, Turkey · 2010 – 2021

- High Honour graduate on academic scholarship · Top 0.2% nationally in the YKS 2021 entrance exam among 3M+ candidates · **2nd Place**, Mathématiques Sans Frontières.

Research & Projects

Parallel MRI Reconstruction: Kernel Geometry under Anisotropic Sampling

- Gained 0.65 dB and matched, with 11 calibration taps, the reconstruction quality the standard rule needed 19 taps to reach, by deriving the kernel-support rule from acquisition physics rather than an arbitrary index-space convention. Validated across 8,947 paired reconstructions against an exact null control.
- Showed that a published kernel-shape equivalence breaks down on anisotropic acquisition grids, and exposed a fault in the reference pipeline that invalidated 27 of 36 cells in the comparison it was built to support.

Decoding Imagined Speech from Non-Invasive EEG

- Reached 46% cross-validated accuracy against a 25% chance baseline. Built a kernel SVM with sequential minimal optimisation, Bayesian logistic regression and an EEGNet-class convolutional network from first principles in NumPy, hand-deriving every gradient with no autograd. A cross-corpus control isolated the near-chance ceiling as a property of the dataset rather than the pipeline.

Autonomous Jenga-Playing Robot

- Hierarchical control stack coupling a learned discrete block-selection policy to an optimisation-based motion planner, bridging reinforcement learning and continuous trajectory synthesis for contact-rich manipulation.

Further coursework projects: medical visual question answering by low-rank adaptation of a vision-language model · convolutional classifier built from scratch to discriminate real from synthetic faces · FPGA colour-sensing joint controller in VHDL · ARM Cortex-M0+ interactive embedded system · Tesla coil and JFET high-impedance probe for non-contact electric field measurement.

Experience

Volina AI · volina.ai · *Co-Founder* Ankara, Turkey · 2025 – present

- AI voice-agent platform for Turkish health-tourism dental clinics, re-engaging dormant international leads with agents that profile callers in real time and adapt tone, objection handling and pacing mid-conversation.
- Delivered five clinic pilots covering roughly 40,000 leads, with per-pilot conversion rates of 1.2% to 3.8%. Raised $25K at $833K pre-money, with KVKK and GDPR-compliant call handling and consent capture built into the platform.

Boğaziçi University MIMLAB · *Volunteer Researcher* İstanbul, Turkey · 2025 – 2026

- Multimodal EEG preprocessing and classification: adaptive filtering against physiological reference channels, artefact evaluation, ERP extraction from n-back tasks, and machine-learning models for cognitive workload state classification.

ASELSAN · *Radar Systems Engineering Intern* Ankara, Turkey · July – August 2024

- Signal-processing optimisation for airborne radar systems: CFAR detection, pulse compression and pulse integration, improving detection accuracy and range through SNR enhancement.

Turkish Entrepreneurship Foundation (GirVak) · *Fellow* İstanbul, Turkey · 2023 – 2026

- One of 28 Fellows selected from 154,558 applicants to Türkiye's leading entrepreneurship programme; previously one of 100 Challengers selected from 149,766 applicants.

Bilkent University Physics Laboratory · *Teaching Assistant* Ankara, Turkey · 2023 – 2024

- Teaching assistant for PHYS102 (General Physics II) and laboratory tutor for PHYS101 (General Physics I).

Skills, Languages & Interests

Programming: Python (NumPy, SciPy, PyTorch, scikit-learn, MNE, pandas), MATLAB, C, C++, JavaScript/TypeScript, VHDL
Methods: Neural signal processing (EEG, ECoG, ERP, spike trains), Bayesian inference, Monte Carlo simulation, deep learning, parallel MRI reconstruction, LLM fine-tuning (LoRA) · **Tools:** Git, Docker, LTspice, LaTeX
Languages: Turkish (native) · English (IELTS Academic 7.5) · French (DELF B2) · **Interests:** Stand-up comedy, mechanical watches, quantitative horse-racing form modelling